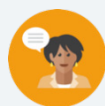


5.11 Writing and Solving Systems of Equations for Real-World Contexts

Lesson Introduction

 Benchmarks	MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.		 Learning Targets	- I can write and solve systems of equations in a real-world context. - I can determine the constraints to a system of equations in a real-world context. - I can interpret solutions as viable or non-viable in a context.
 Benchmark Coherence	Building On:			Working Toward:
	MA.8.AR.4.1 Given a system of two linear equations and a specified set of possible solutions, determine which ordered pairs satisfy the system of linear equations.		MA.912.AR.9.2 Given a mathematical or real-world context, solve a system consisting of a two-variable, linear equation, and a non-linear equation algebraically or graphically.	
	MA.8.AR.4.2 Given a system of two linear equations represented graphically on the same coordinate plane, determine whether there is one solution, no solution, or infinitely many solutions.		MA.912.AR.9.3 Given a mathematical or real-world context, solve a system consisting of two-variable linear or non-linear equations algebraically or graphically.	
	MA.8.AR.4.3 Given a mathematical or real-world context, solve systems of two linear equations by graphing.			
 Essential Questions	- How can I write and solve systems of equations in a real-world context? - How do you determine the constraints to a system of equations from a real-world context?			
 Lesson Narrative	In this lesson, students apply what they have learned throughout this unit to write and solve systems of equations in real-world contexts. Students will be able to solve their written system of equations using the method of their choice.			
 Component Summary	5.11.1 Warm-Up (~5 min)	Learning strategies checklist activity (<i>SE p. 284</i>)	5.11.3 Collaboration (15-20 min)	Interpreting solution in context (<i>SE p. 285</i>)
	5.11.2 Guided Practice (10-15 min)	Practice writing systems of equations (<i>SE p. 284</i>)	5.11.4 Your Turn! (10-15 min)	Independent practice activity on writing and solving systems of equations for real-world contexts (<i>SE p. 287</i>)
	5.11.5 Wrap-Up (~5 min)	Learning target reflection activity (<i>SE p. 287</i>)		
 Resources	Glossary		Materials	
	Key Terms: N/A		(Optional) Multiple colored highlighters	
	Additional Terms: N/A			
	Additional Preparation N/A			

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may attach the variable to the incorrect coefficient or use a constant as a coefficient.	- This lesson is comprised completely of real-world problems. Consider your struggling readers when rolling this lesson out. Look for ways to highlight word problem strategies and thinking out loud when reading a problem. - The lesson appears to be short because of the number of problems but they are filled with rich word problems that require a process of decoding. Consider pacing accordingly.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Three Reads This lesson is comprised completely of real-world problems. For this lesson, consider leveraging the Three Reads routine. This routine supports reading comprehension of problems and meta-awareness of mathematical language.	MA.K12.MTR.7.1: Real-World Contexts This lesson consists solely of real-world problems. During this lesson, students will connect mathematical concepts to everyday experiences. These problems will provide opportunities to create models, both concrete and abstract, and perform investigations.
 Differentiate Instruction	Consider using visual aids and multiple colors to highlight key information from written text in the problem to mathematical translation into an expression. Use different colors to highlight key words and phrases with their corresponding translations.	
Lesson Notes:		

5.11.1 Warm-Up: Learning Strategies

Component Overview: Students will mark the actions they will choose to do today and explain which strategy demonstrates a growth mindset.



5.11.1 Warm-Up: Learning Strategies

1. Learning is not something that just happens, it's something you *do*.
 - a. Place a checkmark next to the actions you will choose to do today.
Answers will vary.
 - ☐ I will review my notes and my work.
 - ☐ I will ask questions when I need help.
 - ☐ I will set and maintain a high standard for myself.
 - ☐ I will ask for help if I'm not sure how to get started.
 - ☐ I will remember that my brain grows when I make mistakes.
 - b. Explain which, if any, strategy you picked demonstrates a growth mindset.
Answers will vary.



Consider having students do this on their own and then share their answers with 1-2 different students. Then engage the class in a polling protocol where you call out the action and see who selected that one, eliciting information from the students who voted about why they selected it and how it demonstrates a growth mindset.

5.11.2 Guided Practice: Writing Systems of Equations

Component Overview: Students will practice writing systems of equations from a real-world context. Students will need a partner for a discussion at the end of this activity.



5.11.2 Guided Practice: Writing Systems of Equations

- Scientists from NASA have focused their efforts to compare the amount of carbon dioxide (CO_2) emitted from different forms of travel.

Vehicle	CO_2 Emissions From Starting the Engine, in Kilograms (kg)	CO_2 Emissions per Mile Traveled, in kg
Car	1	0.25
Jet	2	0.14

- Write a system of equations to represent the total amount of emissions, C , from each vehicle from the time the engine is started when traveling a certain number of miles, m .

$$C = 0.25m + 1$$

$$C = 0.14m + 2$$

- Solve the system using the method of your choice.

$$C = 3.\overline{27}$$

$$m = 9.\overline{09}$$

- At what approximate distance will the jet and the car emit the same amount of CO_2 ?

The jet and the car will emit the same amount of CO_2 after traveling about 9 miles.



Students should be familiar with the Three Reads process from this unit. Consider calling on a student to represent the first read, a different student to represent the second read, and a different student to represent the third read. Allow each student to lead the class through each read, facilitating the discussion of each component.



- Which method for solving this system did you choose?
- Why did you choose that method?

5.11.2 Guided Practice: Writing Systems of Equations (continued)

- d. Shayla and Firenze discussed the constraints that exist in this situation.

Shayla said, "There are no constraints on m because the vehicles can travel any number of miles, including parts of miles. However, c must be positive because carbon dioxide is either emitted or not."

Firenze replied, "I agree with you about the carbon dioxide, but you also can't travel a negative number of miles, so the variable, m , cannot have any negative values either."

Represent the constraints in this context using inequalities.

$$c \geq 0$$

$$m \geq 0$$

- e. Firenze concluded from the context that jets are better for the environment because they emit less CO_2 . Discuss with your partner why this may not be an accurate interpretation.



Engage the class in a discussion around their answers and thinking to question 1d. These are two common misconceptions that come up with constraints with systems of equations. This can provide a safe way to have students learn about misconceptions they may have without self-identifying as having that same error.



- *How do you think that Firenze drew this conclusion?*
- *How would you help him better understand?*

5.11.3 Collaboration: Interpreting Solutions in Context

Component Overview: Students will collaborate on two real-world scenarios to interpret the solution of a system in context. Students will need a partner for this activity.



5.11.3 Collaboration: Interpreting Solutions in Context

1. The equation $t = 71.03x + 3290.51$ represents the relationship between the total average cost of tuition each year, t , in dollars, for a full-time student attending a public university, and years since 2011, x . The total average cost of tuition for a full-time student attending a private university was \$17,710.25 in 2011, and has increased by an average of \$919.25 per year since then.

- a. Write a second equation to represent the relationship between total average cost, t , and years since 2011, x , for private university tuition.
 $t = 919.25x + 17710.25$

- b. Using any method, solve the system of equations created with the two equations.
 $t = 2083$
 $x = -17$

- c. In what year did both private and public universities cost the same amount?
1994

- d. What was the average total cost of tuition at that time?
\$2,083



For struggling readers, consider using visual aids and multiple colors to highlight key information from written text in the problem to mathematical translation into an expression. Use different colors to highlight key words and phrases with their corresponding translations.



- What method did you choose to solve this system?
- Why did you choose that method?

5.11.3 Collaboration: Interpreting Solutions in Context (continued)

2. Tyler prepares overnight oats for his breakfast each morning during the work week. He adds chia seeds for added nutrition. He buys the chia seeds and steel-cut oats he uses at a local bulk store. Last month, he purchased 0.8 pounds of chia seeds and 2.5 pounds of steel-cut oats for \$8.82. This month, he bought 0.4 pounds of chia seeds and 3 pounds of steel-cut oats for \$7.79.

- a. Write a system of equations that could be used to find the price per pound for chia seeds and steel-cut oats at the bulk store. Define the variables used.

s = steel cut oats

c = chia seeds

$$0.8c + 2.5s = 8.82$$

$$0.4c + 3s = 7.79$$

- b. Solve the system of equations. Round your answers to the nearest thousandth.

$$c = 4.989$$

$$s = 1.931$$

- c. Discuss with your partner any constraints on the variables in this context. Then, summarize your discussion.

Answers will vary. The value of both variables must be positive. Both variables must be rounded to the nearest hundredth because both represent money.

- d. Based on your work in Part B and discussion in Part C, interpret a viable solution to the problem.

The price of steel cut oats is \$1.93 per pound and the price of chia seeds is \$4.99 per pound.



For this problem, students may struggle to solve because they may not know what to do if they get a non-terminating decimal. If students are struggling to complete the solution, consider pausing to discuss question 2c before finishing question 2b.

5.11.4 Your Turn!

Component Overview: Students will independently write and solve a system of equations represented by a real-world context, including the interpretation of the solution.



5.11.4 Your Turn!

1. The average homeowner in Florida pays \$150 per month on their energy bills. Some homeowners choose to install solar panels to reduce their monthly energy costs to an average of \$55 per month. The cost of purchasing and installing solar panels is \$13,250 on average.

- a. Write a system of equations that can be used to determine how many months it would take for the total cost of energy bills without solar panels to cost the same as an energy bill with solar panels. Define the variables used.

$$C = 150m$$

$$C = 55m + 13,250$$

- b. Solve the system of equations. Round your answers to the nearest thousandth.

$$c = 20,921.053$$

$$m = 139.474$$

- c. Interpret the solution.

The total cost of using the solar panels is the same as not using solar panels after about 139.5 months. For both options, the cost is about \$20,921 at this time.



Consider using this time to work with individual students or small groups, based on those who were struggling during the guided practice.



- How can you verify that your solution in question 1b is correct?
- Why is it important to check that before doing question 1c?

5.11.5 Wrap-Up: Reflection

Component Overview: Students will indicate how they feel about each learning target from this unit.



5.11.5 Wrap-Up: Reflection

1. Draw an **X** on each line to indicate how you feel about each learning target.

Answers will vary.

- a. I can write and solve systems of equations in a real-world context.

I need help.

I can't get started.

I'm getting there, but

I need more practice.

I understand this

and I feel confident.



- b. I can interpret solutions as viable or non-viable in a given context.

I need help.

I can't get started.

I'm getting there, but

I need more practice.

I understand this

and I feel confident.



Visit unit resources for digital assessment tools like EdgeXL, Test Yourself!, and more for meaningful opportunities to measure student progress on systems throughout the unit. Alternatively, consider project-based applications in which students must construct real-world examples of their systems in action.